5G Mobile Networks Explained

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Introduction to 5G

5G stands for the fifth generation of mobile network technology. It is the latest widely deployed standard for cellular communication, designed to offer much faster data speeds, extremely low delay, and the ability to connect a huge number of devices at the same time. 5G is defined by the International Telecommunication Union under the IMT-2020 standard and by the 3GPP standards body. It is expected to support not only faster smartphones but also smart factories, self-driving vehicles, remote surgery and large-scale Internet of Things networks.

Generations of Mobile Networks

1G, launched in the 1980s, used analogue signals for voice calls only. 2G, from the early 1990s, introduced digital voice, SMS and the GSM standard. 3G, from the early 2000s, brought mobile internet and video calling. 4G LTE, from around 2009 and 2010, enabled fast mobile broadband, HD video streaming and app-based services such as ride-hailing. 5G, commercially launched in 2019 in South Korea and the United States, takes this much further.

How Is 5G Faster Than 4G?

5G is faster than 4G for several technical reasons. First, it uses a wider range of radio spectrum, including new mid-band frequencies and very high millimetre-wave (mmWave) bands from 24 GHz up, which carry large amounts of data with wide channels. Second, it uses massive MIMO, which means base stations with dozens or hundreds of antennas that send and receive many data streams at once. Third, it uses beamforming, which focuses the radio signal in a narrow beam directly toward each user instead of spreading it in all directions. Fourth, it has a more efficient air interface called 5G New Radio (NR). In theory 5G can reach peak speeds of up to 20 Gbps, compared with about 1 Gbps for 4G, and real-world 5G speeds are commonly several times those of 4G, often hundreds of Mbps. 5G also reduces latency, the delay between sending a request and getting a response, to as little as 1 millisecond in ideal conditions, compared with roughly 30 to 50 milliseconds for 4G. In addition, 5G can support up to about one million devices per square kilometre.

Frequency Bands and Network Architecture

Low-band 5G (below 1 GHz) covers wide areas but is not much faster than 4G. Mid-band 5G (1 to 6 GHz, including the 3.3 GHz band) balances speed and coverage and is the main band used in India. High-band mmWave offers the fastest speeds over short range. Non-standalone (NSA) 5G uses the existing 4G core network, while standalone (SA) 5G has its own new 5G core and unlocks all features including network slicing. Network slicing creates multiple virtual networks on the same physical infrastructure, each tailored to a use, such as one for hospitals and one for entertainment. Edge computing places servers near users to reduce delay further.

Three Service Categories

5G has three main use categories. Enhanced Mobile Broadband (eMBB) gives faster internet for streaming and downloads. Ultra-Reliable Low-Latency Communications (URLLC) serves time-critical tasks. Massive Machine-Type Communications (mMTC) connects vast numbers of low-power sensors.

Applications That Need Low Latency

Two applications that need 5G's very low latency are remote surgery, where a surgeon controls robotic instruments from a distance and any delay could be dangerous, and autonomous or connected vehicles, which must exchange information with each other and with road infrastructure within milliseconds to avoid accidents. Other examples are industrial automation with robots, drone control, cloud gaming, augmented and virtual reality, and smart grid protection.

5G in India

India's 5G spectrum auction was held in July and August 2022 and raised about Rs. 1.5 lakh crore in bids. 5G services were launched on 1 October 2022 at the India Mobile Congress in New Delhi by Prime Minister Narendra Modi. Reliance Jio launched True5G, Bharti Airtel launched 5G Plus, and Chennai was among the first cities to receive 5G, together with Delhi, Mumbai and Kolkata. Within about two years, 5G coverage reached nearly every district of the country, and India became one of the fastest 5G rollouts in the world. Indian companies are building 5G equipment, and the government released the Bharat 6G Vision in March 2023. Private 5G networks are being tried at ports, mines and factories.

Challenges and Future

5G needs many more small towers because high-frequency signals travel short distances and are blocked by walls. It requires heavy investment, consumes more power and raises cybersecurity concerns. Future work focuses on 5G-Advanced and 6G, expected around 2030, with even higher speeds, AI-native networks and satellite integration.

Quick Facts

- 5G stands for fifth generation of mobile network technology.
- 5G is faster than 4G because of wider spectrum, massive MIMO, beamforming and 5G NR; peak theoretical speed is up to 20 Gbps.
- Remote surgery and autonomous vehicles are applications that need 5G's roughly 1 millisecond latency.
- India launched 5G on 1 October 2022 at the India Mobile Congress.
- Network slicing creates multiple virtual networks on the same physical 5G infrastructure.